Monocular 6-DoF Object Pose Tracking with Learned Depth and Matching Priors

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***Abstract*—Monocular 6-DoF object pose tracking is ill-posed without depth: scale, shape, and motion are entangled, and small per-frame errors accumulate along the trajectory. We revisit the problem as an integration study: we replace MonoTracker's affine-invariant depth prior with metric-scale monocular geometry (MoGe-2 with frame-0 field-of-view frozen across the clip), retain detector-free dense matching (LoFTR / Efficient LoFTR) and promptable object segmentation (SAM 3), and feed the resulting 2D–3D correspondences into the same joint bundle adjustment. On HO3D under SE(3) alignment – the protocol that preserves absolute trajectory scale – this scale-consistent prior reduces MonoTracker's absolute translation error from 53.3 to 16.0 cm ($3.3\times$), while the methods are essentially tied under the scale-absorbing Sim(3) protocol used by the baseline. The gain is therefore an absolute-scale gain rather than a general tracking-accuracy gain: per-frame relative-rotation error increases from 6.37° to 7.99°. We also discuss, as a future direction rather than a detector claim, how rigid-object tracking may expose inconsistencies in generated hand–object video.**

***Index Terms*—Monocular object pose tracking, hand-object interaction, learned depth priors, dense feature matching, AI-generated video analysis.**

I. Introduction

Estimating the six-degree-of-freedom (6-DoF) pose of a rigid object from a monocular RGB video – no depth sensor, no per-instance CAD model, no multi-view rig – is fundamentally ill-posed: translation along the optical axis is indistinguishable from a change in scale, the object's shape is itself unknown, and small per-frame errors accumulate along the trajectory. MonoTracker [1] closes part of this gap by combining monocular depth predictions with dense feature matching and a joint pose–scale optimization, but accuracy still degrades sharply on hand-object benchmarks such as HO3D [2].

We argue that the missing ingredient for *absolute-scale* monocular tracking is a stronger geometric prior together with reliable correspondences. We pair the metric-scale geometry of MoGe-2 [3] – whose predicted intrinsics we *freeze* at frame 0 to keep the depth scale consistent across the clip – with detector-free matchers (LoFTR [4], Efficient LoFTR [5]) and a promptable segmenter (SAM 3 [6]), and feed the resulting 2D–3D correspondences into MonoTracker's joint pose–disparity bundle adjustment. On HO3D under SE(3) alignment – the protocol that reports a trajectory in absolute units – this outperforms MonoTracker by $3.3\times$ in absolute translation error, while remaining competitive under the scale-absorbing Sim(3) protocol used in the baseline's own paper. This framing is important: once Sim(3) absorbs global scale, our method does not improve local tracking accuracy and has slightly worse relative rotation.

As a secondary observation, current generative video models are optimized primarily for plausible humans and treat the rigid objects in those humans' hands only implicitly, so the object's shape, scale, and texture often drift across frames. A tracker that *assumes* the object is rigid is, by construction, sensitive to this drift. Applied unchanged to one such generated hand-object clip, our tracker recovers a per-frame motion trace that contrasts cleanly with that on a real clip; because this is only a matched pair of clips, we treat it as a future direction rather than a contribution.

Our **contributions** are: *(i)* a controlled MonoTracker-style monocular 6-DoF tracker that swaps in metric geometry with a frame-0 FoV freeze while retaining the same optimizer; and *(ii)* a 52-sequence HO3D evaluation showing a $3.3\times$ gain in SE(3) absolute translation from scale consistency, together with the counterbalancing Sim(3) parity and relative-rotation regression.

II. Related Work

Object pose tracking. Depth-based 6-DoF trackers (BundleTrack [7], BundleSDF [8], FoundationPose [9]) establish the joint-pose-and-geometry template that monocular pipelines try to recover from learned priors alone but require a depth sensor. The closest monocular work is MonoTracker [1], which combines affine-invariant DepthAnything-v2 [10] depth with dense matching and a joint pose–scale BA, leaving two scale parameters free per frame. We replace the depth backbone with metric-scale MoGe-2 [3], freeze its predicted intrinsics across the sequence, and swap mask propagation for SAM 3 [6], keeping the BA formulation so that gains are attributable to the priors.

Geometric foundation models and dense matching. DUSt3R [11] regresses per-pixel point maps; MoGe-2 [3] adds metric scale and a predicted FoV, which we use directly as a per-frame prior. For correspondences we use detector-free matchers [4], [5]; DINOv2-based RoMa [12] is a stronger alternative we do not adopt due to its much higher cost.

Analysis of AI-generated video. Existing methods use appearance-domain cues [13] – frequency artifacts, blending boundaries, biological signals [14] – or spatio-temporal inconsistency [15]. We do not propose a detector; we note only that current generators target plausible humans and treat rigid objects in those humans' hands only implicitly, so a tracker

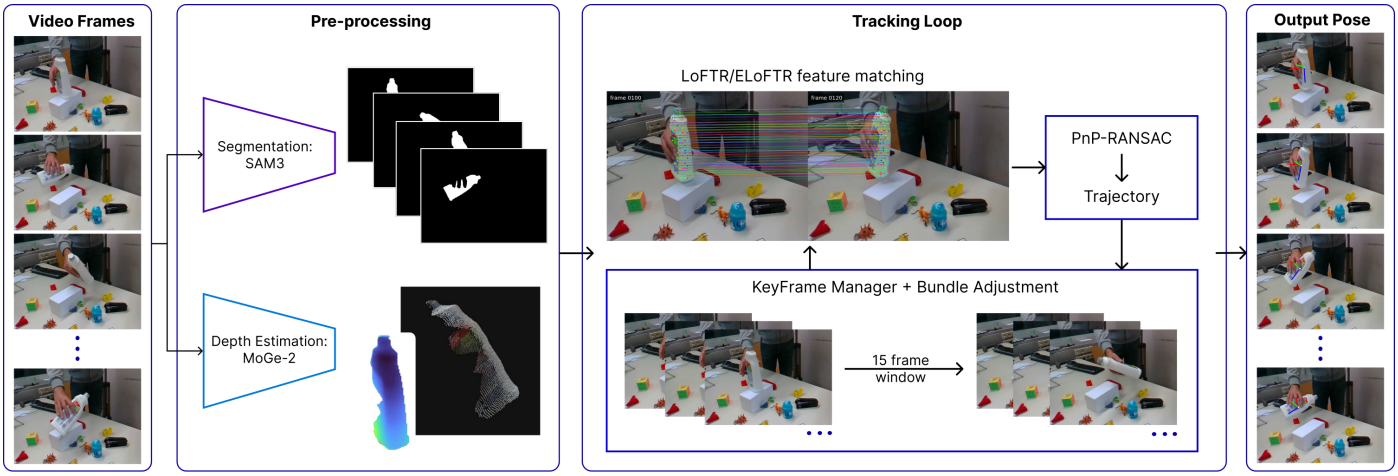


Fig. 1. Overview of our monocular 6-DoF object pose tracker. Per-frame priors – MoGe-2 metric depth with the predicted field of view frozen at frame 0, and a SAM 3 object mask – are combined with dense LoFTR/E-LoFTR correspondences and a PnP-RANSAC initialization, then refined by a joint pose–disparity bundle adjustment over a sliding window of up to $M = 15$ rotation-diverse keyframes with residual weights $(\lambda_{3D}, \lambda_{\text{reproj}}, \lambda_{\text{disp}}) = (5, 100, 12)$. The output is an absolute-scale per-frame trajectory, with no 3D model of the object required at inference.

that assumes object rigidity may be sensitive to that side-effect (Sec. V).

III. Method

A. Overview

Given a monocular RGB video $\{I_i\}_{i=0}^{N-1}$ with shared intrinsics K and an object of interest specified in frame 0 by a SAM 3 prompt, our goal is to recover the per-frame 6-DoF object pose $T_i = [R_i \mid \mathbf{t}_i] \in \mathrm{SE}(3)$ that maps a canonical object frame (anchored at I_0) into camera i. The pipeline has three stages: (i) per-frame geometric priors from a metric monocular geometry model and a promptable segmentation model; (ii) dense cross-frame correspondences from a detector-free matcher; and (iii) a joint pose–disparity bundle adjustment with windowed keyframe edges that produces the final trajectory.

B. Per-Frame Geometric Priors with Fixed Intrinsics

For each frame I_i, we run MoGe-2 [3] to obtain a dense metric depth map $D_i \in \mathbf{R}^{H\times W}$ together with predicted camera intrinsics. MoGe-2 predicts both depth and a per-image field of view, and if invoked naively per frame this leads to a slightly different camera model – and therefore a slightly different depth scale – at every timestep. To make the predicted depth *temporally consistent* we adopt a simple but important modification: we run MoGe-2 once on frame 0, recover its predicted horizontal field of view $\phi_x^{(0)}$ from the returned normalized focal length via $\phi_x^{(0)} = 2\arctan(0.5/f_{x,\text{norm}}^{(0)})$, and then condition every subsequent inference call on the frozen value $\phi_x^{(0)}$. With the camera model held fixed across the sequence, any residual inter-frame depth-scale variation could in principle be absorbed by a single affine depth correction (a_i, b_i) per frame, jointly optimized in stage (iii). In our final configuration these parameters are clamped (Sec. III-D); the formulation is retained because it follows MonoTracker's pose–scale optimization and lets us toggle the disparity residual cleanly for analysis.

Object masks M_i are produced by SAM 3 [6] from one or more object-category text prompts at frame 0 and propagated through the video. When a prompt returns multiple candidates, we retain the candidate with the largest frame-0 area; we accept it only if that area is at least 200 pixels. The matcher and the geometry term only operate on pixels with $M_i = 1$, which both speeds up matching and prevents background geometry from leaking into the pose estimate.

C. Dense Correspondences and Pose Initialization

For each pair of consecutive frames (I_{i-1}, I_i), we obtain semi-dense correspondences $\{(u_{i-1}^k, u_i^k)\}_k$ from a transformer-based matcher; our implementation supports LoFTR [4] and Efficient LoFTR [5] as drop-in front-ends. Matches falling outside M_{i-1} or M_i are discarded, and pairs whose predicted depths exceed the matched in-mask mean by more than 3σ are rejected to suppress occlusion-edge outliers in D_i.

The surviving 2D correspondences are lifted to 3D via $\mathbf{p}_{i-1}^k = D_{i-1}(u_{i-1}^k) \cdot K^{-1}\tilde{u}_{i-1}^k$ and fed to a RANSAC PnP solver [16] to produce an initial relative pose $T_{i-1\to i}$. This initial pose is composed with T_{i-1} to seed T_i for the optimization that follows; only inlier matches are kept.

D. Joint Pose–Disparity Bundle Adjustment

The core of our pipeline is a joint optimization over poses and per-frame depth scale–shift parameters, following MonoTracker's formulation but with a different prior (metric depth instead of affine-invariant depth). For each frame we keep a pair of parameters (a_i, b_i) that refine the depth predicted by MoGe-2,

$$\hat{z}_i(u) \;=\; a_i D_i(u) + b_i, \tag{1}$$

so that the refined depth $\hat{z}_i$ corrects any residual scale and shift left over after the fixed-intrinsics MoGe-2 call. For a match (u_i^k, u_j^k) between an edge of keyframes $(i,j) \in \mathcal{E}$, let $\mathbf{p}_i^k$ be the 3D point lifted from frame i using $\hat{z}_i$ and let $\hat{u}_j^k = \pi(KT_{i\to j}\mathbf{p}_i^k)$ be its projection into frame j. We minimize a sum of three residuals robustified by the Huber norm [17]:

$$\mathcal{L} = \sum_{(i,j)\in\mathcal{E}} \sum_k \big[\, \lambda_{\text{3D}} \left\|\mathbf{p}_i^k - T_{i\to j}^{-1}\mathbf{p}_j^k\right\|_h + \lambda_{\text{reproj}} \left\|\hat{u}_j^k - u_j^k\right\|_h + \lambda_{\text{disp}} \left|\hat{z}_j (u_j^k)^{-1} - \hat{z}_i^{\to j}(u_j^k)^{-1}\right|_h \big], \quad (2)$$

where $\mathcal{E}$ is the set of frame pairs in the current bundle, $\hat{z}_i^{\to j}$ denotes the depth of $\mathbf{p}_i^k$ after transforming into frame j, and $\|\cdot\|_h$ is the Huber norm. The three terms penalize, respectively, 3D inconsistency of corresponding points, 2D reprojection error, and disparity mismatch in inverse-depth space; we use weights $(\lambda_{\text{3D}}, \lambda_{\text{reproj}}, \lambda_{\text{disp}}) = (5, 100, 12)$ throughout. The objective is minimized with a Ceres-based Levenberg–Marquardt solver [18].

a) Windowed keyframe BA.: After each two-frame local optimization, every Δ frames (we use $\Delta = 5$) we trigger a windowed bundle adjustment in which the current frame is paired with up to $M = 15$ previously selected keyframes, using the rotation-diverse keyframe-selection criterion of MonoTracker [1]. This step refines the most recent poses against earlier views and is what prevents long-horizon drift.

b) Fixing the depth scale.: Because MoGe-2 already produces *metric* depth, the depth scale–shift parameters (a_i, b_i) have little to absorb in practice. We clamp $(a_i, b_i) = (1, 0)$ throughout (the `FIX_DEPTH_SCALE` option), removing two degrees of freedom per frame; this configuration is used for all numbers reported in Sec. IV.

IV. Pose Tracking on HO3D

A. Setup

We evaluate on the HO3D [2] hand–object interaction benchmark, the evaluation target used by MonoTracker [1]. We start from the 55 HO3D training sequences and exclude three sequences (GPMF10, MDF13, SM5) for which SAM 3 propagation still produced an empty mask within the evaluated window after re-generation attempts; no sequence is excluded based on pose-tracking accuracy. For the remaining 52 sequences, we take 251 frames starting at frame 100 with stride 2. This is a broader and denser-sampled subset than the 9-sequence stride-5 subset reported in the MonoTracker paper; it makes the comparison harder (more frames, more sequences) but more representative of the full benchmark. We compare three configurations on identical inputs (same RGB, same SAM 3 masks, same evaluator):

- **MonoTracker** – LoFTR matching with DepthAnything-v2 depth, as released by the original authors.
- **Ours-LoFTR** – our pipeline with LoFTR matching, MoGe-2 metric depth, frame-0 field-of-view freeze, and the disparity-down-weighted joint BA.
- **Ours-ELoFTR** – identical to Ours-LoFTR except the matcher is swapped for Efficient LoFTR [5].

We report two protocols. **SE(3) frame-0 alignment** composes a single rigid transform that aligns the first predicted pose to the first ground-truth pose; the recovered trajectory is then evaluated as-is, so absolute scale errors are not absorbed. **Sim(3) alignment** additionally fits a global scale to the predicted translation trajectory via Umeyama's closed-form similarity alignment [19][1]. We report standard SE(3)/Sim(3) metrics – absolute translation error, recall at 5/10 cm, and AUC of translation error. ADD is the mean distance between object mesh vertices transformed by predicted and ground-truth poses, while ADD-S uses nearest-neighbor distances for symmetric or ambiguous objects [20], [21]; AUC ADD-S@0.1 m integrates the recall curve up to 0.1 m. We additionally report the gauge-free *per-frame relative-rotation error*, the mean angle between consecutive ground-truth and predicted relative rotations, which is independent of any alignment.

B. Results

Table I aggregates results over all 52 sequences. The dominant headline is on SE(3) translation: MonoTracker achieves 53.3 cm mean translation error, while our integrated configuration replaces DepthAnything-v2 with MoGe-2, freezes the predicted intrinsics, clamps the depth correction, and uses residual weights $(5, 100, 12)$, reducing the error to 16.0 cm – a $3.3\times$ improvement – without changing the BA formulation. The same effect is visible in the metric-scale-sensitive recalls and AUCs: SE(3) recall@10 cm rises from 0.120 to 0.483, AUC of ADD-S@0.1 m rises from 0.109 to 0.398.

The reason is direct. DepthAnything-v2 outputs an affine-invariant prediction that lives in arbitrary units; frame-0 SE(3) alignment can rotate and translate the predicted trajectory but cannot rescale it, so the baseline accumulates a trajectory whose absolute extent is wrong by an order of magnitude. MoGe-2 returns metric depth, and the frame-0 field-of-view freeze keeps that depth scale more consistent across the sequence; the resulting trajectory is substantially closer to the correct absolute scale before alignment. Sim(3) alignment is exactly the protocol that absorbs this scale gap. Under Sim(3) the recovered global scale factor (pred→gt) is 0.086 for MonoTracker and 0.122 for Ours-LoFTR – our prediction is still off the true metric scale by roughly $8\times$, but it is markedly closer to unity than the relative-depth baseline.

Under Sim(3) the three methods are essentially tied on absolute translation ($\approx$ 5 cm) and on AUC ADD-S@0.1 m (0.467/0.473/0.463): once trajectory scale is absorbed, the per-frame correspondence quality is what matters and the matchers are comparable. This is consistent with the spirit of MonoTracker's published numbers and confirms that the gains we report on SE(3) come from the metric-depth prior rather than from a different optimizer or matcher.

[1]This is the protocol used in MonoTracker's HO3D table; it makes the comparison scale-invariant.

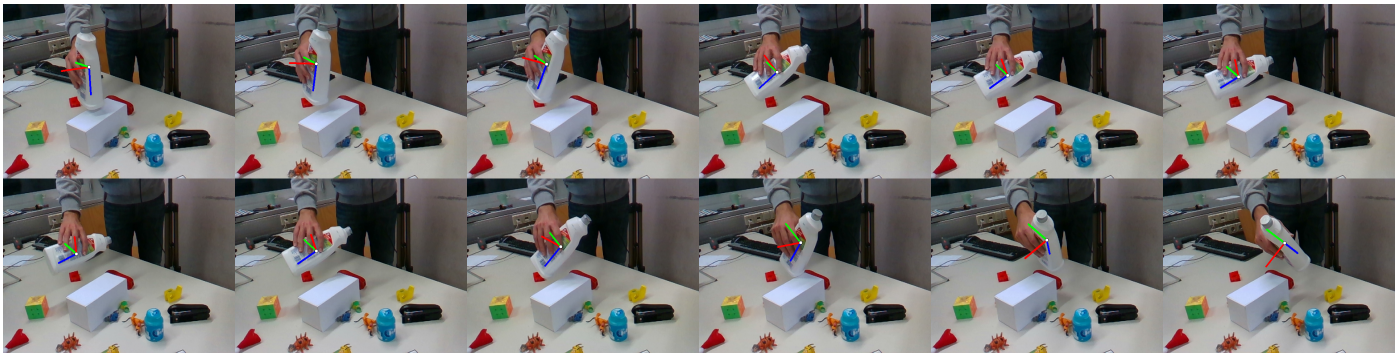


Fig. 2. Qualitative pose recovery on HO3D ABF11. Twelve frames sampled at 20-frame intervals from frame 100 to frame 320, top-left to bottom-right; X red, Y green, Z blue. Axis origins are anchored at the SAM 3 mask center for visibility; orientations come directly from the predicted pose. The axes stay locked to the bottle across the full sampled range despite substantial hand motion.

a) Net takeaway.: On Sim(3), MonoTracker edges ours by a small margin (5.19 vs. 5.39 cm; 0.661 vs. 0.638 recall@5 cm), since once a global scale is fit both methods recover comparable local geometry. But this view is conditional on access to ground-truth scale at evaluation time, which a real deployment never has. In the SE(3) setting – the protocol that actually reports trajectories in absolute units – ours wins on every metric: 3.3× lower translation error (16.0 vs. 53.3 cm), ∼ 4× recall@10 cm, ∼ 3.6× AUC of ADD-S@0.1 m.

A second observation is that all three methods exhibit large absolute rotation error (≈ 57° under SE(3), ≈ 116° under Sim(3) where Umeyama's translation-only fit does not realign rotation), and the per-frame relative-rotation error is slightly *worse* for our pipeline (7.99°/7.97° versus 6.37° for MonoTracker). We attribute this to MoGe-2's depth being smoother than DepthAnything-v2's along the object's silhouette: smoother lifted 3D reduces the leverage of the 3D consistency term in the BA on small in-plane rotations, while greatly stabilizing the long-horizon translation. This is a per-frame versus long-horizon trade-off, and the trade is favorable only when absolute translation scale is the primary requirement.

Efficient LoFTR is within noise of LoFTR on every metric and is faster in our implementation; we therefore recommend it as the default front-end.

b) Qualitative pose recovery.: Fig. 2 overlays the predicted object axes on twelve frames of HO3D ABF11 sampled at 20-frame intervals from frame 100 to frame 320 (about 7 s of real-time motion at the native 30 fps capture). The axis origin is placed at the center of the SAM 3 mask in each frame so it tracks the object as it is carried by the hand; the axis *orientations* are taken from the predicted pose with no further alignment. Across the twelve frames the axes remain locked to the bottle through translations, in-plane rotations, and out-of-plane tilts, with no visible drift in orientation – a qualitative check that complements the aggregate metrics in Table I.

TABLE I
AGGREGATE HO3D RESULTS OVER 52 SEQUENCES. BEST PER METRIC IN **BOLD**. OURS* = OUR PIPELINE WITH LOFTR; OURS† = OUR PIPELINE WITH EFFICIENT LOFTR.

Metric	MonoTracker	Ours*	Ours†
SE(3) frame-0 alignment			
abs. trans. err. (cm) ↓	53.30	**16.01**	16.77
abs. rot. err. (°) ↓	57.03	**56.33**	57.62
recall @ 5 cm ↑	0.080	**0.232**	0.231
recall @ 10 cm ↑	0.120	**0.483**	0.471
AUC trans. @ 5 cm ↑	0.046	0.111	**0.112**
AUC ADD-S @ 0.1 m ↑	0.109	**0.398**	0.383
Sim(3) alignment (paper protocol)			
recovered scale (pred→gt)	0.086	**0.122**	0.114
abs. trans. err. (cm) ↓	**5.19**	5.39	5.48
recall @ 5 cm ↑	**0.661**	0.638	0.615
recall @ 10 cm ↑	**0.907**	0.895	0.894
AUC ADD-S @ 0.1 m ↑	0.467	**0.473**	0.463
Per-frame (gauge-free)			
rel. rot. err. (°) ↓	**6.37**	7.99	7.97

V. LIMITATIONS AND REPRODUCIBILITY

Generalization. Our quantitative evaluation is limited to HO3D, whose videos have calibrated intrinsics, hand-held YCB-style objects, and short-range manipulation. The method should transfer most directly to other monocular RGB videos with reliable object masks, enough visual texture for dense matching, and a reasonably fixed camera model. We expect degraded performance for heavy occlusion, strong specular or textureless objects, severe motion blur, object symmetries that make rotation ambiguous, or segmentation failures; these are exactly the cases where correspondence quality or the rigidity assumption breaks.

Ablations and repeatability. The full 52-sequence table is a controlled single run for each method, using identical RGB frames, masks, evaluator, and optimizer settings. After depth and correspondence caching, the remaining stochastic element is mainly PnP-RANSAC; we therefore avoid claiming statistical significance for the small Sim(3) gaps. We did not complete a full multi-seed study or a 52-sequence per-frame-FoV ablation. On GPMF13, clamping the MoGe-2 depth scale improved SE(3) translation from 22.6 to 19.0 cm, and the final weights $(5, 100, 12)$ further improved it to 17.1 cm. In a separate 251-frame GPMF13 diagnostic with identical regenerated masks and clamped scale, freezing FoV yielded 10.92 cm versus 11.04 cm with per-frame FoV, only a 1.1%

gain; this suggests that the metric prior, rather than the freeze alone, drives most of the headline improvement.

Runtime and release. On a single NVIDIA RTX 5090 GPU, MoGe-2 preprocessing took about $15.7\,\mathrm{s}$ per 251-frame HO3D sequence in our logs; the subsequent tracking loop averaged $0.59\,\mathrm{s/frame}$ with LoFTR and $0.37\,\mathrm{s/frame}$ with Efficient LoFTR. We will release the evaluation scripts, sequence list, masks, and cached trajectory files to make the table reproducible.

Generated-video note. We also observed that, on one generated hand–object clip, the tracker produced sustained per-frame object motion that was absent from a matched real HO3D clip. This is only an anecdotal $n = 1$ observation: we have not measured false-alarm rate, generator dependence, object-category dependence, or sensitivity to prompt and motion regime. We therefore do not treat it as a detector, but as a future direction for studying whether rigid-object motion consistency can complement appearance-domain generated-video analysis [13]–[15].

VI. Conclusion

We presented a monocular 6-DoF object pose tracker that replaces the missing depth channel with two learned priors – metric monocular geometry from MoGe-2 with frozen frame-0 intrinsics, and dense LoFTR / Efficient LoFTR correspondences – and feeds the result into a MonoTracker-style joint pose–disparity bundle adjustment. On HO3D we outperform MonoTracker by $3.3\times$ on SE(3) absolute translation error and by roughly $4\times$ on metric-scale recall, while remaining competitive under the scale-absorbing Sim(3) protocol used by the baseline's authors. The result should be read narrowly: the gain comes from absolute-scale consistency supplied by the metric depth prior and frame-0 FoV freeze, while local relative rotation remains worse than MonoTracker. Future work should add multi-seed variance, isolate the FoV freeze over the full benchmark, evaluate beyond HO3D, and test whether rigid-object motion consistency is useful for generated-video analysis across a real corpus.

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